Date:

Christoffel Symbol of first kind

$$\Gamma_{ijk} = \frac{1}{2}\left[\frac{\partial}{\partial x^i}(g_{jk}) + \frac{\partial}{\partial x^j}(g_{ki}) - \frac{\partial}{\partial x^k}(g_{ij})\right]$$

$$= \frac{1}{2}\left[g_{jk,i} + g_{ki,j} - g_{ij,k}\right)$$

Transformation law for Christoffel Symbol of first kind.

$$\tilde{\Gamma}_{ijk} = \Gamma_{rst}\frac{\partial x^r}{\partial \tilde{x}^i}\frac{\partial x^s}{\partial \tilde{x}^j}\frac{\partial x^t}{\partial \tilde{x}^k} + \frac{\partial^2 x^r}{\partial \tilde{x}^i \partial \tilde{x}^j}\frac{\partial x^s}{\partial \tilde{x}^k}$$

Christoffel Symbol of Type II

$$\Gamma^i_{jk} = g^{ir}\Gamma_{jkr}$$

$$\Gamma^i_{jk} = \frac{g^{ir}}{2}\left[-g_{jk,r} + g_{kr,j} + g_{rj,k}\right].$$

Transformation law for Christoffel Symbol of ~~first~~ II kind.

$$\Gamma^i_{jk} = \Gamma^r_{st}\frac{\partial \tilde{x}^i}{\partial x^r}\frac{\partial x^s}{\partial \tilde{x}^j}\frac{\partial x^t}{\partial \tilde{x}^k} + \frac{\partial^2 x^r}{\partial \tilde{x}^j \partial \tilde{x}^k}\frac{\partial \tilde{x}^i}{\partial x^r}$$

Covariant:

$$T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^t_{ik} T_t$$

Contravariant:

$$T^i_{;k} = \frac{\partial T^i}{\partial x^k} - \Gamma^i_{tk} T^t$$

Date:

Simple Lorentz transformation

$$\tilde{x}^0 = \frac{x^0 - \beta x'}{\sqrt{1-\beta^2}}, \quad \tilde{x}' = \frac{-\beta x^0 - x'}{\sqrt{1-\beta^2}}$$

$$\tilde{x}^2 = x^2, \quad \tilde{x}^3 = x^3$$

Relativity velocity $\beta = \frac{v}{c}$.

$a^2 - b^2 = 1$

$$\Lambda = \begin{bmatrix} a & b & 0 & 0 \\ b & a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \Lambda^{-1} = \begin{bmatrix} a & -b & 0 & 0 \\ -b & a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 1/\sqrt{1-\beta^2} & -\beta/\sqrt{1-\beta^2} & 0 & 0 \\ \beta/\sqrt{1-\beta^2} & 1/\sqrt{1-\beta^2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Length Contraction:

$$\Delta\tilde{x}' = \frac{1}{\sqrt{1-\beta^2}} \Delta x' \Rightarrow \Delta x' = \sqrt{1-\beta^2}\,\Delta\tilde{x}'$$

Date:

Absolute Tensor

$$T^i_{;j} \frac{dx^j}{dt}$$

$$\frac{\delta T^i}{\delta t} = \frac{dT^i}{dt} + \Gamma^i_{rs} T^s \frac{dx^r}{dt}$$

Torsion Tensor:

$$T^\alpha_{\beta\gamma} = T^\alpha_{\beta\gamma} - T^\alpha_{\gamma\beta}$$

Acceleration in General Coordinates

$$\vec{r} = \vec{r}(x^i(t))$$

$$\vec{a} = (a^i) = \frac{d}{dt}\frac{dx^i}{dt} = \frac{d^2x^i}{dt^2}$$

Instantaneous acceleration

$$a = \sqrt{\delta_{ij} a^i a^j}$$

Formula for acceleration

$$= \sqrt{|g_{ij} a^i a^j|}$$

Date:

Generalization of $\frac{d}{dt}\frac{dx^i}{dt}$

$$\vec{a} = (a^i) = \frac{d^2x^i}{dt^2} + \Gamma^i_{rs}\frac{dx^r}{dt}\frac{dx^s}{dt}$$

Curvature :-

$$k(s) = \sqrt{\delta_{ij}\frac{d^2x^i}{ds^2}\frac{d^2x^j}{ds^2}}$$

$$s(t) = \int_a^t \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}\,dt$$

Generalized formula for Curvature

$$k(s) = \sqrt{g_{ij}\,b^i b^j}$$

$$b^i = \frac{\delta}{ds}\left(\frac{dx^i}{ds}\right) = \frac{d^2x^i}{ds^2} + \Gamma^i_{rs}\frac{dx^r}{ds}\frac{dx^s}{ds}$$

$$s(t) = \int_a^t \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}\,dt$$

Minkowski spacetime :-

$$ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$

Date:

Non-Comoving

$$u = (\gamma c, \gamma v^x, \gamma v^y, \gamma v^z)$$

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

$$T^{ij} = \rho \gamma^2 \begin{vmatrix} c & cv^x & cv^y & cv^z \\ cv^x & (v^x)^2 & v^x v^y & v^x v^z \\ cv^y & v^y v^x & (v^y)^2 & v^y v^z \\ cv^z & v^z v^x & v^z v^y & (v^z)^2 \end{vmatrix}$$

Conservation eq. for energy momentum tensor

$$\frac{\partial T^{ij}}{\partial x^j} = 0$$

$i = 0$, $i = t$

$$\frac{\partial T^{tj}}{\partial x^j} = \frac{\partial T^{tt}}{c \partial t} + \frac{\partial T^{tx}}{\partial x} + \frac{\partial T^{ty}}{\partial y} + \frac{\partial T^{tz}}{\partial z}$$

$$= \frac{1}{c} \frac{\partial}{\partial t} (\rho \gamma^2 c^2) + \frac{\partial}{\partial x} (\rho \gamma^2 v^x)$$

$$+ \frac{\partial}{\partial y} (\rho \gamma^2 v^y) + \frac{\partial}{\partial y} (\rho \gamma^2 v^z)$$

Date:

Geodesic: / Normal / Riemannian

$y^i = sp^i$

Einstein tensor:

$$G^i_j = R^i_j - \frac{1}{2}\delta^i_j R$$

Energy momentum tensor

T^{ij}

Covariant derivative of energy momentum tensor

$$\nabla_i T^{ij} = 0$$

Stress energy tensor

$T^{ij} = \rho u^i u^j$ u^i (4) vapor velocity

Energy-Momentum tensor of Dust in Minkowski Spacetime (flat Space)

Co-moving.

$u^i = (c, 0, 0, 0)$

Two events

$x_1^0, x_1^1, x_1^2, x_1^3 \longrightarrow A$

$x_2^0, x_2^1, x_2^2, x_2^3 \longrightarrow B$

$$\tilde{x}_1^1 = \frac{-\beta x_1^0 - x_1^1}{\sqrt{1-\beta^2}}$$

$$\tilde{x}_2^1 = \frac{-\beta x_2^0 - x_2^1}{\sqrt{1-\beta^2}}$$

$$\Delta\tilde{x}^1 = \frac{1}{\sqrt{1-\beta^2}}\left[-\beta\Delta x^0 + \Delta x^1\right]$$

$$\Delta\tilde{x}^1 = \frac{1\,\Delta x^1}{\sqrt{1-\beta^2}}$$

Time Dilation (Increase)

$$\Delta x^0 = \frac{1}{\sqrt{1-\beta^2}}\Delta\tilde{x}^0 \Rightarrow \Delta\tilde{x}^0\sqrt{1-\beta^2} = \Delta x^0$$

$$x_1^0 = \frac{\tilde{x}_1^0 - \beta\tilde{x}_1^1}{\sqrt{1-\beta^2}} \quad , \quad x_2^0 = \frac{\tilde{x}_2^0 - \beta\tilde{x}_2^1}{\sqrt{1-\beta^2}}$$

$$\Delta x^0 = \frac{\Delta\tilde{x}^0 + \beta\Delta\tilde{x}^1}{\sqrt{1-\beta^2}}$$

$$\Delta x^0 = \frac{\Delta x^0}{\sqrt{1-\beta^2}}$$

Date:

Simple Lor

$\tilde{x}^0 = \frac{x^0 - \beta}{\sqrt{1-}}$

$\tilde{x}^1 = x^2$

Boldsymbol

$A =$ | a b | / | b a | / | a a | / | a a |

$A =$

Date:

Einstein field Eq.

$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

poission law:-

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} = 4\pi G\rho$$

General relativistic analogue of poission eq.

$$\left(R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R\right)_{;\beta} = 0$$

$$\Downarrow$$

$$R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R = -K T_{\alpha\beta}$$

$$K = \frac{8\pi G}{c^4}$$

Maxwell Field Eq.'s

$$\nabla \cdot E = \frac{\rho}{\varepsilon_0}$$

$$\nabla \cdot B = 0$$

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \times B = \mu_0 j + \frac{1}{c^2}\frac{\partial E}{\partial t}$$

Date:

Non-Comoving

$u^i = \gamma$

$T^{ij} = (\rho + \frac{P}{c^2}) u^i u^j - P g^{ij}$

$T^{00} = (\rho + \frac{P}{c^2}) \gamma^2 c^2 - P$

$= \rho c^2 + P\gamma^2 - P$

$T^{00} = \rho c^2 + (\gamma^2 - 1) P$

(In Non-diagonal pressure is zero).

$T^{01} = (\rho + \frac{P}{c^2}) u^0 u^1 - 0$

$= (\rho + \frac{P}{c^2})(\gamma c)(\gamma v^x)$

$T^{01} = \gamma^2 c v^x (\rho + \frac{P}{c^2})$

$T^{02} = v^y$

$T^{03} = v^z$ $\quad \delta'' = -1$

$T^{11} = (\rho + \frac{P}{c^2})(\gamma v^x)(\gamma v^x) \oplus P(g'')$

$= (\rho + \frac{P}{c^2})(\gamma^2 v^x)^2 + P$

$T^{12} = T^{13}$

Date:

✱ Gradient

$$\nabla\phi = \text{grad}\,\phi = \frac{\partial \phi}{\partial x^i}\hat{e}_i$$

✱ divergence

$$\text{div}(A^i) = A^i_{;i}$$

$$= A^i_{,i} + \Gamma^i_{ik} A^k$$

✱ In covariant form

$$\text{div}(A_i) = A_{i;i} = A_{i,i} - \Gamma^s_{ii} A_s$$

✱ Curl

$$(\nabla \times A)_i = \epsilon_{ijk} A_{k;j}$$

In covariant derivative.

$$(\nabla \times A)_i = \epsilon_{ijk} A_{k;j}$$

$$A_{k;j} = A_{k,j} - \Gamma^s_{kj} A_s$$

✱ Laplacian:-

$$\nabla^2\phi = \nabla \cdot \nabla\phi$$

$$\nabla\phi = \frac{\partial \phi}{\partial x^i}\hat{e}_i$$

$$\nabla \cdot \nabla\phi = \left(\frac{\partial \phi}{\partial x^i}\right)_{;i}$$

div (grad ϕ).

Date:

Type B R_{abac} $b<c$.

$$n_b = 3\,{}^nC_3 = \frac{n(n-1)(n-2)}{2}$$

Type C R_{abcd} $a<b<c<d$

$$n_c = 2\,{}^nC_4 = \frac{n(n-1)(n-2)(n-3)}{12}$$

Corollary.

$$R_{1212} = R_{2121} = -R_{1221} = -R_{2112}$$

Ricci Tensor

$$R_{ab} = R^c_{acb}$$

$$\boxed{R_{ab} = R_{ba}}$$

$$R_{ab} = g^{ab} R_{ab} = R^a_a$$

Riemannian Curvature

$$K = K(x^i, u^i, v^i) = \frac{R_{ijkl} u^i v^j u^k v^l}{G_{pqrs} u^p v^q u^r v^s}$$

$$G_{pqrs} = g_{pr} g_{qs} - g_{ps} g_{qr}$$

Date:

Four velocity:-

$$\tilde{v}_t = c = v_t$$

$$\tilde{v}_x = \frac{v_x - v}{1 - \frac{v_x v}{c^2}} \qquad \tilde{v}_y = \frac{v_y \sqrt{1-\beta^2}}{1 - v_x}$$

$$\tilde{v}_z = \frac{v_z \sqrt{1-\beta^2}}{1 - \frac{v_x v}{c^2}}$$

Four acceleration:-

$$\tilde{a}_x = \frac{a_x (1-\beta^2)^{3/2}}{\left(1 - \frac{v_x v}{c^2}\right)^3}$$

$$\tilde{a}_y = \frac{a_y + [a_x v_y - v_x a_y] v/c^2 \; (1-\beta^2)}{\left(1 - \frac{v_x v}{c^2}\right)^3}$$

$$\tilde{a}_z = \frac{a_z + [a_x v_z - v_x a_z] v/c^2 \; (1-\beta^2)}{\left(1 - \frac{v_x v}{c^2}\right)^3}$$

Proper time:-

$$\tau = \frac{s}{c} = \frac{1}{c} \int_{t_0}^{t} \sqrt{g_{ij} \frac{dx^i dx^j}{du \, du}} \, du$$

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}} \qquad v < c.$$

Date:

Charge density:

$$\rho_0 = \frac{Ne}{l_0^3} \quad \text{in } S'$$

In S

$$\rho = \frac{Ne}{l_0^3 \sqrt{1-v^2/c^2}}$$

$$\rho = \frac{\rho_0}{\sqrt{1-v^2/c^2}}$$

Current density:

in S' (no current frame)

In S

$$\vec{j} = \rho \vec{v}$$

$$\vec{j} = \frac{\rho_0 \vec{v}}{\sqrt{1-v^2/c^2}} \qquad \rho = \frac{\rho_0}{\sqrt{1-v^2/c^2}}$$

In component form

$$j_x = \frac{\rho_0 v_x}{\sqrt{1-v^2/c^2}} \qquad j_y = \frac{\rho_0 v_y}{\sqrt{1-v^2/c^2}} \qquad j_z = \frac{\rho_0 v_z}{\sqrt{1-v^2/c^2}}$$

Date:

Riemann tensor

$$(V_{i;j})_{;k} - V_{i;jk}$$

$$R^i_{jkl} = \frac{\partial \Gamma^i_{jl}}{\partial x^k} - \frac{\partial \Gamma^i_{jk}}{\partial x^l} + \Gamma^n_{jl}\Gamma^i_{nk} - \Gamma^n_{jk}\Gamma^i_{nl}$$

properties of Riemann Tensor

$$R^i_{jkl} = -R^i_{jlk}$$

$$R_{ijkl} = -R_{ijlk} \quad \text{(second skew symmetric)}$$

$$R_{ijkl} = -R_{jikl} \quad \text{(first skew symmetric)}$$

$$R_{ijkl} = R_{klij}$$

$$R_{ijkl} + R_{iklj} + R_{iljk} = 0 \quad \text{(Bianchi's Identity)}$$

$$R_{aacd} = R_{iicd} = -R_{iicd} \Rightarrow R_{iicd} = 0$$

Type: A R_{abab} $a < b$.

$$n(A) = {}^nC_2 = \frac{n(n-1)}{2!}$$

Date:

$\Rightarrow T^{(i)} = T^{ij} = \rho u^i u^j = \rho u^i u^0 = \rho c \cdot c$

$T^{00} = \rho c^2$

$$T^{(i)} = \begin{bmatrix} \rho c^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$v^i = \frac{dx^i}{dt} = (ct, x, y, z)$

$= (c, v^x, v^y, v^z)$

$\frac{d\tau}{dt} = \sqrt{1 - v^2/c^2} = \frac{1}{\sqrt{1 - v^2/c^2}} = \gamma \quad \Rightarrow \frac{dt}{d\tau}$

$u^i = \frac{d}{d\tau}(ct, x, y, z)$

$= \frac{dt}{d\tau} \frac{d}{dt}(ct, x, y, z)$

$= \gamma (c, v^x, v^y, v^z)$

$v = 0$

$u^i = c(1, 0, 0, 0)$

Energy momentum tensor of perfect fluid

$$T^{ij} = \left(\rho + \frac{P}{c^2}\right) u^i u^j - P g^{ij}$$

Co-moving

$$u^i = (c, 0, 0, 0)$$

$$T^{ij} = \begin{bmatrix} c & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{bmatrix}$$

$$T^{ij} = \left(\rho + \frac{P}{c^2}\right) u^0 u^0 - P g^{00}$$

$$= \left(\rho + \frac{P}{c^2}\right) c^2 - P$$

$$T^{00} = \rho c^2$$

$$T^{11} = P$$

$$T^{22} = P$$

$$T^{33} = P$$

Date:

Proper time velocity

$$u^i = \frac{dx^i}{d\tau}$$

Proper time acceleration

$$b^i = \frac{du^i}{d\tau} = \frac{d^2u^i}{d\tau^2}$$

Rest Mass

$$\hat{m} = \frac{m}{\sqrt{1-\vec{v}^2/c^2}}$$

4-momentum:

$$p^i = (p^0, \vec{p}) = \hat{m}\vec{v}_i$$

Lorentz force $(F_0, \vec{F})$

$$\vec{F} = \frac{d}{dt}(\hat{m}v_0) = \frac{m\vec{a}}{\sqrt{1-v^2/c^2}}$$

$$= \frac{d}{dt}\left(\frac{m\vec{v}}{\sqrt{1-v^2/c^2}}\right)$$

Minkowski force

$$K = \frac{f_i}{\sqrt{1-\vec{v}^2/c^2}}$$

Date:

Short expressions for components

$$F_{ij} = \varepsilon_{ijk} B^k \qquad f_{0i} = -\frac{E_i}{c}$$

$$F^{ij} = \varepsilon^{ijk} B_k \qquad f_{0i} = E_i/c$$

Maxwell field eq. in tensor form

$$F_{\alpha\beta;\gamma} + F_{\beta\gamma;\alpha} + F_{\gamma\alpha;\beta} = 0$$

$$F^{\alpha\beta}{}_{;\beta} = \mu_0 J^\alpha$$

In flat space

$$F^{\alpha\beta}{}_{,\beta} = \mu_0 J^\alpha$$

$$F_{\alpha\beta,\gamma} + F_{\beta\gamma,\alpha} + F_{\gamma\alpha,\beta} = 0$$

Electromagnetic EMT (Energy momentum tensor)

$$T^\alpha_\beta = \frac{1}{4}\delta^\alpha_\beta \left(F_{\mu\nu}F^{\mu\nu}\right) - F_{\nu\beta}F^{\nu\alpha}$$

or

$$T^{\alpha\beta} = \frac{1}{4}\delta^{\alpha\beta}\left(F_{\mu\nu}F^{\mu\nu}\right) - F^{\alpha\mu}F^\beta_\mu$$

* Trace (Electromagnetic EMT)

$$T^\alpha_\alpha = T^0_0 + T^1_1 + T^2_2 + T^3_3$$

Date:

$$\frac{\partial}{\partial t}\left(\frac{\phi}{c}\right) + \nabla \cdot \vec{A}$$

$$\frac{1}{c^2}\frac{\partial \phi}{\partial t} + \nabla \cdot \vec{A} = 0$$

★ Electromagnetic field tensor

$$F^{\alpha\beta} = \nabla^\alpha A^\beta - \nabla^\beta A^\alpha$$

$$A^\alpha = \left(\frac{\phi}{c}, \vec{A}\right)$$

- In flat space

$$F^{\alpha\beta} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

- $F_{\alpha\beta} = g_{\alpha\gamma} F^{\gamma\delta} g_{\delta\beta}$

$$F_{\alpha\beta} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

Date:

Conservation eq. for four vector current density

$$\nabla_\alpha J^\alpha = J^\alpha{}_{;\alpha} = 0$$

In Special Relativity

$$J^\alpha{}_{;\alpha} = J^0{}_{,0} + J^1{}_{,1} + J^2{}_{,2} + J^3{}_{,3}$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0$$

Electromagnetic four potential

$$A^\mu = \left(\frac{\varphi}{c}, \vec{A}\right)$$

$$U = q\varphi$$
$$\vec{\omega} = q\vec{A}$$

Magnetic and electric field in term

$$\vec{B} = \nabla \times \vec{A} \quad , \quad \vec{E} = -\nabla\varphi - \frac{\partial \vec{A}}{\partial t}$$

Four potential conservation

$$\nabla_\alpha A^\alpha = 0 \quad , \quad A^\alpha{}_{;\alpha} = 0$$

In flat space

$$A^\alpha{}_{;\alpha} = A^0{}_{,0} + A^1{}_{,1} + A^2{}_{,2} + A^3{}_{,3}$$

Date:

Properties of $F^{\mu\nu}$

- Anti symmetry $F^{\mu\nu} = -F^{\nu\mu}$
- Trace : $F = F^{\mu}{}_{\mu} = 0$
- It has Six Independent components
- $F_{\mu\nu} F^{\mu\nu} = 2\left(B^2 - E^2/c^2\right)$

Date:

Mass: In S' $= m_0$.

In S frame

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

$$\rho = \frac{\rho_0}{\sqrt{1 - v^2/c^2}}$$

$$\frac{\rho}{m} = \frac{\rho_0}{m_0} \Rightarrow \rho = \frac{\rho_0 m}{m_0}$$

$$\vec{j} = \rho \vec{u}$$

$$= \left(\frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}\right) \rho$$

$$\vec{j} = \frac{\rho_0 m \vec{u}}{m_0}$$

$$\vec{j} = \left(\frac{\rho_0}{m_0}\right) \rho$$

Four current:

$$J^\mu = (\rho c, \vec{j})$$

$$\vec{j} = j_x \hat{i} + j_y \hat{j} + j_z \hat{k}$$

$$J_\mu = (\rho c, -\vec{j})$$

Conservation of Current

$$\nabla_\mu J^\mu =$$

In Special

$$J^\mu{}_{;\mu} =$$

$$\frac{\partial \rho}{\partial t} +$$

Ele

$$J_\mu = g_{\mu\nu} J^\nu$$

$$= g_{00} J^0 + g_{01} J^1 + g_{02} J^2 + g_{03} J^3$$

$$= \rho c - J^1 - J^2 - J^3$$

$$= \rho c - \vec{j}$$